Road2Career-AI-Powered Personalized Career Mentor and Roadmap Generator

A PROJECT REPORT

Submitted by

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Under the Guidance of

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to

the APJ Abdul Kalam Technological University
in partial fulfillment of the requirements for the award of the degree
of
Master of Computer Applications



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DECLARATION

I undersigned, hereby declare that the project report "Road2Career – AI-Powered Personalised Career Mentor and Roadmap Generator" submitted in partial fulfilment of the requirements for the award of the degree of Master of Computer Applications of APJ Abdul Kalam Technological University, Kerala, is a bona fide work carried out by me under the supervision of Mrs.Anjali Shankar. This submission represents my ideas in my own words, and wherever the ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to the principles of academic honesty and integrity and have not misrepresented or fabricated any data, idea, fact, or source in this submission. I understand that any violation of the above may result in disciplinary action by the institute and/or the University and may also lead to further action by the sources whose work has not been properly cited or from whom permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma, or similar title from any other university.

Place : KALADY Signature

Date : GLADIA VARGHESE

ADI SHANKARA INSTITUTE OF ENGINEERING AND TECHNOLOGY

VIDYA BHARATHI NAGAR, MATTOOR, KALADY

DEPARTMENT OF COMPUTER APPLICATIONS

CERTIFICATE

This is to certify that the report entitled Title "**Road2Career-AI-Powered Personalized Career Mentor and Roadmap Generator**" submitted by **GLADIA VARGHESE ASI25MCA-2027** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Master of Computer Applications, is a Bonafide record of the project work carried out by him/her under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Ms.Anjali Sankar
Project Guide
Department of Computer Applications

Dr. Vincy Devi V K
Head of the department
Department of Computer Applications

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ABSTRACT

Choosing a suitable career is a major challenge for students due to the wide range of career opportunities available and the difficulty of identifying the skills and qualifications required for a particular career path. Existing career guidance platforms often provide general recommendations that may not fully address individual interests, skills, and career goals. Road2Career is an AI-powered web application designed to provide personalised career guidance, identify skill gaps, and generate structured career roadmaps to help students make informed career decisions. The system provides a web-based interface through which students can register, log in, manage their profiles, select career goals, and assess their existing skills. Based on the student's profile, interests, and skill assessment, the system identifies the skills that need improvement and provides personalised career recommendations. Artificial intelligence techniques support the generation of customised career roadmaps, while a career readiness score helps students understand their current level of preparation for their chosen career. The application also includes a weekly learning planner, roadmap history, progress tracking, and a career timeline to organise learning activities and monitor development.

The system is developed using React for the frontend, Node.js and Express.js for backend services, and PostgreSQL for database management. These technologies support user authentication, profile management, skill tracking, career goal management, and the storage of relevant user and roadmap information. The modular architecture allows the application to organise career guidance activities and provides scope for future improvements in recommendation accuracy and personalisation.

The main objective of Road2Career is to bridge the gap between students' existing skills and the requirements of their desired careers by providing clear, personalised, and actionable guidance. The system aims to help students understand their career options, develop relevant skills, follow structured learning plans, and track their progress towards career readiness. Road2Career serves as a practical foundation for intelligent and personalised career planning and can be further enhanced with advanced recommendation algorithms, real-time career information, and integration with learning platforms.